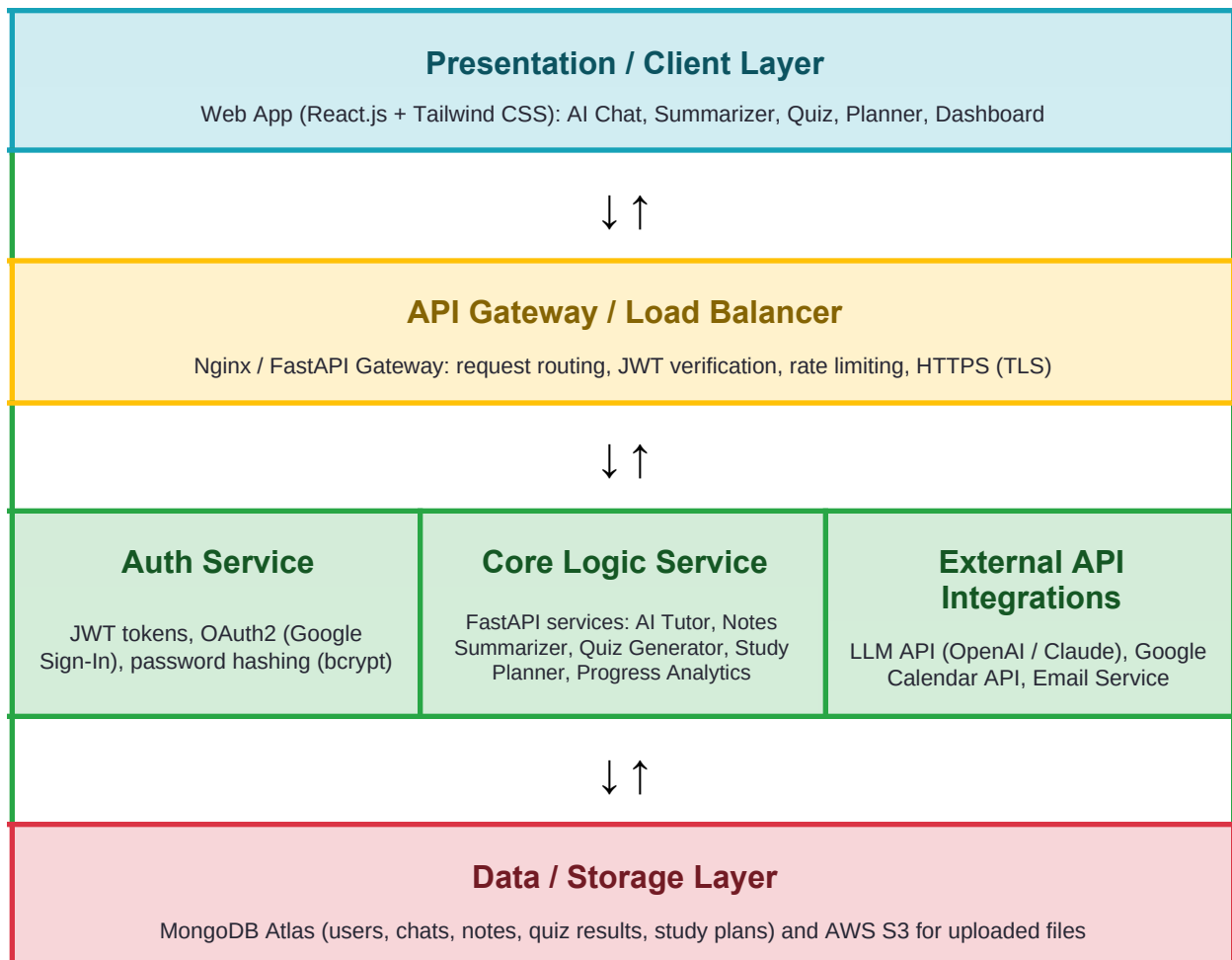


Solution Architecture

Date	30 September 2026
Team ID	SWTID-2026-5807
Project Name	AI-StudyBuddy
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Responsive UI for AI chat, notes upload, quizzes, planner and progress dashboard.	React.js, Tailwind CSS
API Gateway	Single entry point that routes requests, verifies JWT tokens and applies rate limits.	Nginx, FastAPI, HTTPS/TLS
Microservices	Auth, AI Tutor, Summarizer, Quiz Generator, Planner and Analytics services with the core logic.	Python, FastAPI, LLM API, OAuth2/JWT
Database	Stores user profiles, chat history, notes, quiz results and study plans; files in cloud storage.	MongoDB Atlas, AWS S3